NetPerturb: a reproducible Nextflow pipeline for in silico perturbation of single-cell gene regulatory networks from a Seurat object

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Abstract

Background: Predicting how a genetic or pharmacological perturbation affects each cell population in a single-cell RNA-seq (scRNA-seq) dataset is a common question. In silico perturbation tools based on gene regulatory networks (GRNs) can answer it from unperturbed cells, but they are hard to install, scale poorly to large atlases, and return results that are difficult to compare across methods and to share.

Results: We present NetPerturb, a Nextflow pipeline that starts from a processed Seurat object and a list of target genes and returns a single, self-contained HTML report. NetPerturb balances cell identities by downsampling, checks the targets before any network is built, and infers one GRN per identity with GENIE3, scRank or hdWGCNA. It then ranks identities by a target-perturbation score. In parallel, it can run a virtual knockout with scTenifoldKnk, including joint knockouts of several genes, followed by gene set enrichment of the genes the knockout affects. Every stochastic step is seeded from one parameter and every tool runs in a versioned container, so runs are reproducible on HPC and cloud. We benchmarked the pipeline on four acute myeloid leukaemia (AML) cohorts (86 cell identities, 55 drug targets). Target-level scores replicated between cohorts (Spearman 0.88–0.92). Identity-wide offsets tracked the number of cells behind each network; once they were removed, stem cells were not more affected overall, but specific targets were. IL3RA (CD123) and KDM1A (LSD1) consistently acted on stem identities, mainly malignant ones, PSMB5, CDK8 and HMGCR acted mainly on normal stem cells, and mitotic kinases spared stem cells in all

four cohorts. Relative to the other targets, the blast markers KIT, FLT3 and CD33 scored higher in malignant identities than in their normal counterparts.
Conclusions: NetPerturb makes GRN-based in silico perturbation reproducible and scalable, starting from the object most analysts already have. Its scores are best read within a run, through target-specific effects aggregated over identities and, where possible, over cohorts. NetPerturb is freely available at https://github.com/MCBLab/NetPerturb.

Keywords: Single-cell RNA-seq, Gene regulatory network, In silico perturbation, Virtual knockout, Nextflow, Acute myeloid leukaemia, Leukaemic stem cells, Drug targets

Background

Single-cell RNA-seq (scRNA-seq) resolves tissues into cell identities (cell types, clones, or treatment states), and the next question is usually which of those identities depend on a given gene or drug target. Pooled CRISPR screens with single-cell readout answer this experimentally [1], but they are costly and cannot be applied to archived or clinical samples. Computational alternatives therefore perturb a gene regulatory network (GRN) inferred from unperturbed cells and read the effect off the network: CellOracle simulates shifts in cell identity [2], scRank ranks cell types by how much a drug target reshapes their network [3], and scTenifoldKnk performs a virtual knockout and reports the genes whose regulatory context changes [4].

In practice, using these methods on a new dataset takes considerable engineering work. Each method expects its own input format and gene universe, and memory grows quickly with the number of cells and genes. Targets that are not expressed in some identities can end a run late in the analysis. Results from different network inference methods are rarely placed side by side, and many steps are stochastic, so the numbers are hard to reproduce. [TODO: Add 1–2 sentences on the specific gap your lab hit.]

NetPerturb addresses this with a single Nextflow [5] workflow, organised following nf-core practice [6]. Its input is the object most analysts already have, a processed Seurat object [7]. It parallelises network inference and scoring across identities and targets, and it gathers every result, together with the quality control and provenance needed to interpret it, into one portable report. We describe the implementation and then benchmark the rank-score track on four acute myeloid leukaemia (AML) cohorts, asking whether its results replicate between cohorts and how the targets of AML drugs act across the haematopoietic hierarchy.

Implementation

Input and pre-processing

NetPerturb requires four inputs: a Seurat object (`--obj`), the metadata column defining cell identities (`--column`), the species, and a text file of targets (`--target`), one

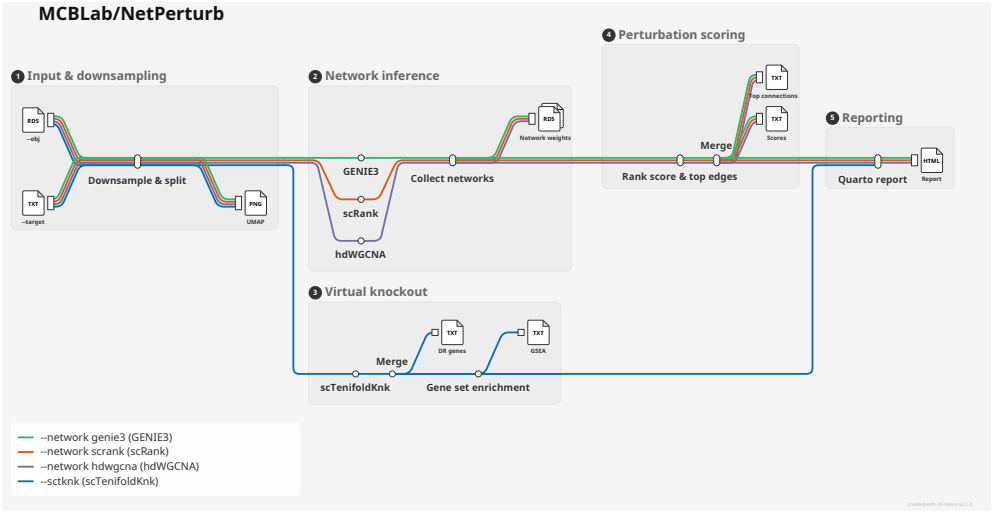


Fig. 1 Overview of NetPerturb. A processed Seurat object and a list of target genes enter `DOWNSAMPLE`, which balances cell identities and checks the targets. The rank-score track (warm lines; one of GENIE3, scRank or hdWGCNA, chosen with `--network`) infers one GRN per identity and scores every identity × target pair. The knockout track (blue; `--sctknk`) builds one wild-type scTenifoldKnk network per identity and runs one virtual knockout per identity × target, optionally followed by fgsea. Both tracks converge on a single self-contained HTML report.

per line (Fig. 1). Genes joined by a semicolon (e.g. `CDK8;CDK19`) are treated as a single combined perturbation. The `DOWNSAMPLE` step randomly keeps at most `--n_cells` cells per identity, so that every identity contributes a comparable number of cells, and drops identities that end up with fewer than `--min_cells` cells. It also builds a shared gene universe from the highly variable genes, transcription factors, drug targets and the requested targets. Each target is checked against the retained cells before any network is built. Genes that are absent from the object or have no counts are removed and recorded, a combined target keeps whichever of its genes pass, and the run stops only when no target passes. Problems that would otherwise end a run hours later are therefore caught at the start and listed in the report.

Rank-score track

For each identity, NetPerturb infers a GRN with one of three interchangeable methods (Table 1). GENIE3 [8] and scRank's own network builder [3] are used as published. hdWGCNA [9], which builds on WGCNA [10], produces an unsigned co-expression network on metacells, so NetPerturb adapts its topological overlap matrix to the input scRank expects in four steps. It restores the sign of each edge from the correlation between the two genes, pads every identity to the same gene set, removes edges below the `--cut_ratio` quantile of absolute weight, and rescales the weights to $[-1, 1]$. Each identity × target pair is then scored with scRank, which compares each network before and after removing the target's edges and returns a perturbation score in either antagonist or agonist mode. The strongest `--top_connections` edges of each target are also

Table 1 Network inference methods available in NetPerturb

Method	Network	Output	Track
GENIE3	Tree-based regression	Directed weights	Rank score
scRank	scRank builder	Directed weights	Rank score
hdWGCNA	Metacell TOM + sign	Signed, sparsified	Rank score
scTenifoldKnk	PCR + tensor denoising	WT vs. KO manifold	Knockout

TOM, topological overlap matrix; PCR, principal component regression; WT, wild type; KO, knockout.

exported. Networks and scores are computed as separate parallel tasks and merged into one table afterwards.

Virtual knockout track

When `--sctknk` is set, a scTenifoldKnk track [4] runs alongside the rank-score track, or on its own. The expensive wild-type network does not depend on the target, so NetPerturb builds it once per identity (`SCTENIFOLDKNK_BUILD`). Each identity × target knockout then runs as a separate task on that stored network (`SCTENIFOLDKNK_KO`). With a single build per identity, one wild-type network can serve any number of knockouts, and the knockout of a combined target can zero the edges of all its genes at once, which the original single-gene interface does not allow. Differential regulation uses the statistic of scTenifoldKnk v1.0.3, which leaves the knocked-out genes out of the reference mean. Because the genes an alignment moves most are the network's hubs whatever the target, each gene is also tested against the same knockout run for random genes of the network (`--sctknk_null`), a target with no outgoing edge is skipped as a no-op, and the rank of the tensor decomposition is a parameter (`--sctknk_td_k`). If a GMT file is supplied, genes are ranked by $\log_2$FC and tested with fgsea [11] against, for example, MSigDB collections [12]. Gene sets are read from local files, so the step also runs on compute nodes without internet access.

Report

The `REPORT` step renders a single self-contained Quarto HTML file ([TODO: Additional file 1: report screenshots]). The report contains: (i) an overview recording the network method, the targets removed by quality control, the number of cells and genes available to each identity's network, and a UMAP of the retained cells; (ii) a heatmap of target expression per identity, so that a low score can be checked against a target that is barely expressed; (iii) a searchable table and a heatmap of perturbation scores; (iv) interactive knockout networks and a table of differentially regulated genes; (v) an enrichment dot plot and table; (vi) data-quality diagnostics for each identity's input and network; and (vii) provenance, with the command line, software versions, parameters and the running time of each step taken from the Nextflow trace.

Reproducibility and deployment

Every tool runs in a versioned Singularity/Docker container [13], so no R dependencies need to be installed. A single `--seed` parameter sets the random seed of every step that uses one: downsampling, GENIE3 (via doRNG), hdWGCNA metacells, scTenifoldKnk and fgsea. scRank's network builder is the one exception, because it sets its own seed internally. Memory and CPU requests are defined per process and can be adjusted in a configuration profile. The pipeline is tested with nf-test at the module and workflow level, and it provides small and dataset-scale test profiles (`-profile test`).

Results

We benchmarked the rank-score track on four AML cohorts processed in the same way (Figs. 2–4), and earlier on one of them with its authors' own malignant/normal labels (Fig. 5). We asked three questions: whether the scores replicate between cohorts, whether stem cells are more affected by the targets of AML drugs than other cells, and which targets distinguish malignant cells from their normal counterparts. The knockout track was not part of this benchmark.

Four AML cohorts processed with one command each

Zeng et al. [14] mapped the cells of 21 published AML scRNA-seq studies onto a reference of normal haematopoiesis, giving every cell one of the same broad cell types whatever its study. We took four of these cohorts [15–18] and called every cell malignant or normal with scMalignantFinder [19]. Its pre-trained pan-cancer model did not separate AML blasts from normal cells (58% agreement with van Galen's labels, AUC 0.59), so we retrained it on the curated labels of the Beneyto-Calabuig, Lasry and Naldini cohorts. On the held-out van Galen cohort the retrained model agreed with the original labels for 81% of cells (AUC 0.88); for the other three cohorts the call is in-sample. [TODO: Confirm that the final calls used this retrained model (3 cohorts, logistic regression).] Each identity combines the atlas's broad cell type with this call, a malignant identity carrying the suffix *-like* (e.g. HSC_MPP-like for malignant cells that resemble haematopoietic stem and multipotent progenitor cells).

We ran NetPerturb on each cohort with GENIE3 networks, antagonist mode, `--n_cells` 3,000 and `--min_cells` 150, and the same 55 targets of drugs used or trialled in AML: 34 single genes (BCL2, FLT3, IDH1, IDH2, MEN1, DNMT1, XPO1, MDM2, KIT, CD33, IL3RA and others) and 21 pairs of them. NetPerturb kept 13 identities in Beneyto-Calabuig, 22 in Lasry, 24 in Naldini and 27 in van Galen, 86 in all (111,592 cells), and dropped another 54 identities below `--min_cells` automatically (Fig. 2A). The cohorts differ in what they contain. Beneyto-Calabuig keeps only three normal identities, all mature myeloid; van Galen is the only cohort with enough lymphocytes, plasma cells and NK cells to score them; and the malignant identities usually reach the 3,000-cell cap while most normal ones hold a few hundred cells. The shared gene universe held 4,232–4,282 genes per cohort, of which 2,133–3,710 had counts in a given identity. Every target passed quality control in every cohort, and only 13 of the 4,730 identity × target pairs had no score, all of them for PLK1, HAVCR2, AURKB, AURKA or DHODH.

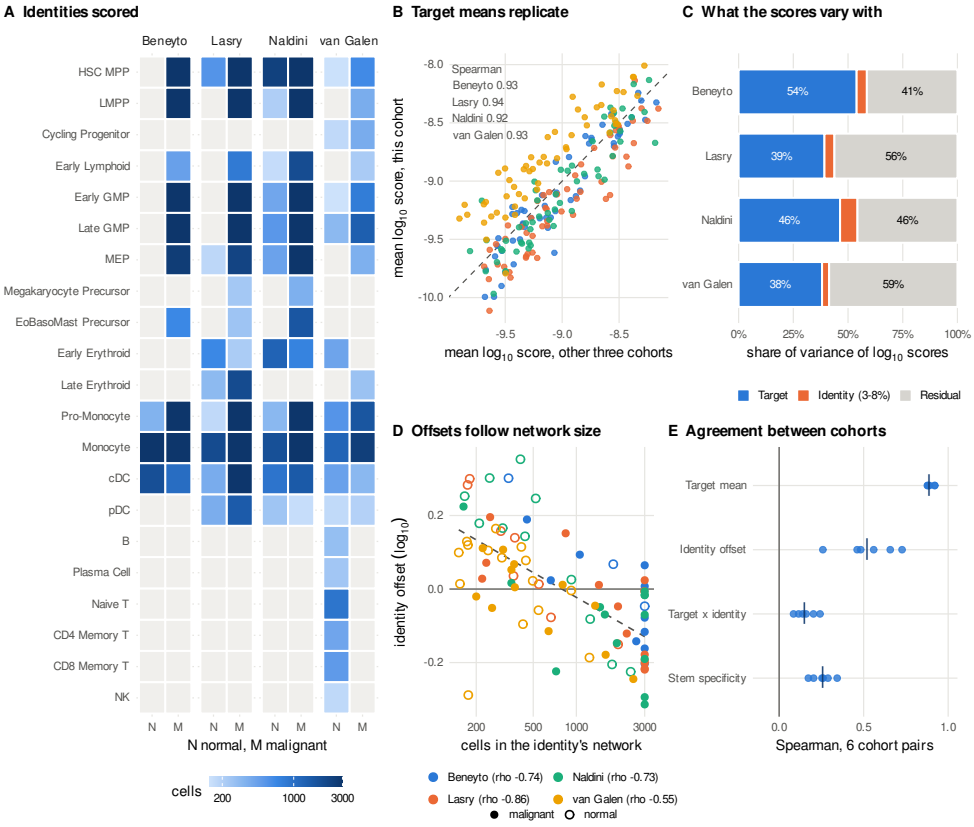


Fig. 2 Four AML cohorts: design, reproducibility and network-size offsets (cohorts of the Zeng et al. [14] atlas; GENIE3 networks, antagonist mode, 55 targets). (A) Identities scored per cohort, normal (N) and malignant (M, scMalignantFinder call), coloured by the number of cells the identity's network was built from; grey, not scored. (B) Mean $\log_{10}$ score of each target in one cohort against its mean over the other three; dashed line, identity. (C) Share of the variance of $\log_{10}$ scores explained by the target, by the identity, and left as the target × identity residual. (D) Identity-wide offset, the identity effect of a target + identity decomposition of $\log_{10}$ scores, against the number of cells in the identity's network; filled, malignant; dashed line, least-squares fit over all cohorts. (E) Spearman correlation over the six pairs of cohorts for the target mean, the identity offset and the target-specific effect (12 identities scored in every cohort), and for stem specificity (Fig. 3B); bars, medians.

Target-level scores replicate across cohorts

Within a cohort, the target explained 38–54% of the variance of $\log_{10}$ scores, the identity 3–8%, and the remaining 41–59% was specific to the target in that identity (Fig. 2C). The target level replicated closely. A target's mean score in one cohort correlated with its mean over the other three with Spearman ρ of 0.92–0.94 (Fig. 2B), and pairs of cohorts agreed with ρ 0.88–0.92 (Fig. 2E). The same held against an earlier run on van Galen's original labels with an older pipeline version ($\rho = 0.84$; Fig. 5). Joint targets scored highest: their median score was 5.5-fold that of single genes, and DNMT1;XPO1, FLT3;DNMT1, DNMT1;IDH2 and BCL2;DNMT1 were in the top ten of every cohort. Among single genes, EIF4A1, DNMT1 and IDH2 ranked highest, and DHODH, AURKA and CDK8 were in the bottom ten in every cohort.

The identity level replicated moderately (ρ 0.26–0.73 for identity offsets over the 12 identities scored in every cohort), and the target-specific part of a single identity × target score, the residual left once target and identity effects are removed, replicated far less (ρ 0.09–0.24; Fig. 2E). A single identity × target score is therefore mostly cohort-specific, and we draw conclusions only from effects aggregated over several identities and over cohorts.

Identity offsets follow network size

Every identity's scores were shifted by an identity-wide offset, which spanned 0.45–0.67 $\log_{10}$ per cohort. The offsets fell as the number of cells behind the identity's network rose, in every cohort (ρ −0.74, −0.86, −0.73 and −0.55 for Beneyto-Calabuig, Lasry, Naldini and van Galen; Fig. 2D), and as the number of genes in the network rose (ρ −0.53 to −0.81). This matters in AML, where malignant identities usually reach the `--n_cells` cap and their normal counterparts do not (median 1,917 against 407 cells). In a model of the offsets on cohort, compartment and malignancy, malignant identities scored 0.13 $\log_{10}$ below normal ones ($P = 1 \times 10^{-4}$). Adding the number of cells to the model raised the variance it explained from 24% to 60% (−0.26 $\log_{10}$ per tenfold more cells, $P = 10^{-12}$), and the malignant difference vanished (−0.03, $P = 0.32$). Stem identities (HSC/MPP and LMPP) then scored slightly below progenitor and mature ones (by 0.08, $P = 0.02$, and 0.07, $P = 0.05$). Stem cells were therefore not more affected by the targets as a whole. The offsets report how large an identity's network is rather than how vulnerable its cells are, so every comparison below uses the target-specific part of each score, from which the offsets are removed.

Targets that consistently act on stem cells

Removing the offsets left the question of which targets act on stem cells more than on the rest of the hierarchy (Fig. 3A). For each target and cohort, we took the target-specific effect averaged over the stem identities minus that averaged over all other identities (Fig. 3B), and tested the mean over cohorts against 10,000 permutations of identity labels within each cohort. Stem specificity was more consistent than chance: 16 targets had the same sign in all four cohorts, 11 towards stem and 5 away from it, against 7.1 expected under permutation ($P = 0.003$; Fig. 3D). Six targets acted more on stem identities in all four cohorts with $P < 0.05$: IL3RA (+0.57 $\log_{10}$,

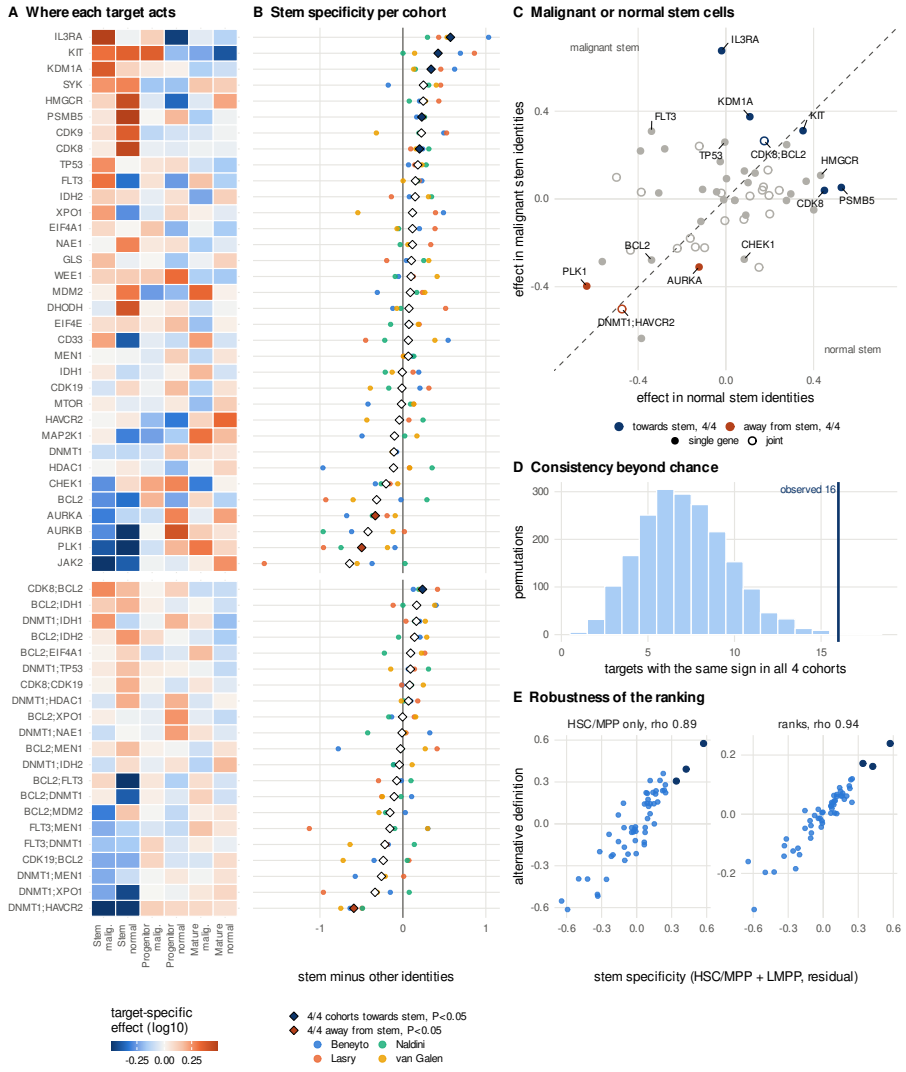


Fig. 3 Targets that act on stem identities. The target-specific effect is the residual of the $\log_{10}$ score once target and identity effects are removed, per cohort. (A) Target-specific effect averaged over the identities of each compartment and then over cohorts; stem, HSC/MPP and LMPP; progenitor, cycling, lymphoid, granulocyte–monocyte, megakaryocyte–erythroid, eosinophil/basophil/mast and early erythroid progenitors; mature, all other identities. (B) Stem specificity, the target-specific effect in stem identities minus that in all other identities, per cohort (dots) and averaged over cohorts (diamonds); filled diamonds, same sign in all four cohorts and $P < 0.05$ against 10,000 permutations of identity labels within each cohort. In A and B, targets are ordered by stem specificity, single genes above joint targets. (C) Target-specific effect in malignant against normal stem identities, averaged over cohorts (normal stem identities are absent from Beneyto-Calabuig); dashed line, identity. (D) Number of targets whose stem specificity has the same sign in all four cohorts, observed (line) and over 2,000 label permutations (histogram). (E) Stem specificity as defined in B against two alternative definitions: HSC/MPP alone as the stem compartment, and the within-identity rank of the target among the 55 in place of the residual; dark points, IL3RA, KIT and KDM1A.

FDR 0.04), KIT (+0.42), KDM1A (+0.34, FDR 0.04), the joint target CDK8;BCL2 (+0.24), PSMB5 (+0.23) and CDK8 (+0.20). HMGCR, TP53, FLT3, DNMT1;IDH1 and MEN1 were positive in all four cohorts without reaching $P < 0.05$. At the other end, the mitotic kinases AURKA (−0.33) and PLK1 (−0.50) acted less on stem identities in all four cohorts ($P < 0.05$), as did CHEK1 (−0.20, $P = 0.10$); as a group, AURKA, AURKB, PLK1, CHEK1 and WEE1 averaged −0.27 ($P = 0.07$). The joint target DNMT1;HAVCR2 had the strongest effect away from stem cells (−0.59, FDR 0.005). The ranking did not depend on how stem cells or effects were defined. It held when the stem compartment was restricted to HSC/MPP ($\rho = 0.89$) and when each target's rank among the 55 within an identity replaced the residual ($\rho = 0.94$; Fig. 3E).

Malignant and normal stem cells respond to different targets

Three of the four cohorts hold both malignant and normal stem identities, so the stem-specific targets could be split by the stem cells they act on (Fig. 3C). IL3RA (CD123), a marker of leukaemic stem cells [20], acted on malignant stem identities (+0.68) and not on normal ones (−0.02), and it was higher in malignant than in normal stem cells in each of the three cohorts. FLT3 behaved the same way (+0.31 against −0.34; higher in malignant stem cells in all three cohorts), as did DNMT1;IDH1 (+0.24 against −0.12) and TP53 (+0.26 against 0.00). KDM1A (LSD1), which sustains leukaemic stem cells [21], leaned the same way (+0.37 against +0.11), although the paired difference was positive in only one of the three cohorts. PSMB5, CDK8 and HMGCR instead acted mostly on normal HSC/MPP (+0.43 to +0.53, against +0.04 to +0.11 in malignant stem cells), and lower in malignant than in normal stem cells in all three cohorts. KIT acted on both (+0.31 and +0.35). The first group is what a target selective for leukaemic stem cells would look like, while the second predicts haematopoietic toxicity. PLK1 and AURKA spared both kinds of stem cell.

Blast markers separate malignant cells from their normal counterparts

Across the 30 matched pairs of a malignant identity and the normal identity of the same broad cell type (Fig. 4), the receptors and antigens that mark AML blasts stood out. KIT, FLT3 and CD33, the blast markers singled out in the single-cohort run below, gained more than the other targets in 23 of the 30 pairs (median +0.27 $\log_{10}$, signed-rank $P = 1.5 \times 10^{-4}$; Fig. 4B). The excess was clearest in progenitors (9 of 10 pairs, median +0.30) and stem cells (3 of 4, +0.37), and weaker in mature identities (11 of 16, +0.16). Individually, BCL2 (+0.34, higher in 70% of pairs, $P = 0.02$) and FLT3 (+0.30, 73%, $P = 0.003$) led (Fig. 4A), and both, with TP53, were higher in malignant identities on average in every cohort. BCL2, the target of venetoclax, was nonetheless not stem-specific relative to the other identities (−0.32; Fig. 3B), although it was higher in malignant than in normal stem cells in all three cohorts with both. At the other end, IDH2 (−0.33, $P = 0.004$), AURKA (−0.29, $P = 0.001$), CDK19, CHEK1, EIF4E and NAE1 were lower in malignant identities ($P < 0.05$). IDH1/2 inhibitors act on a mutant enzyme whose activity an expression network cannot see, so these targets are not expected to follow malignancy. The pair-level detail (Fig. 4C)

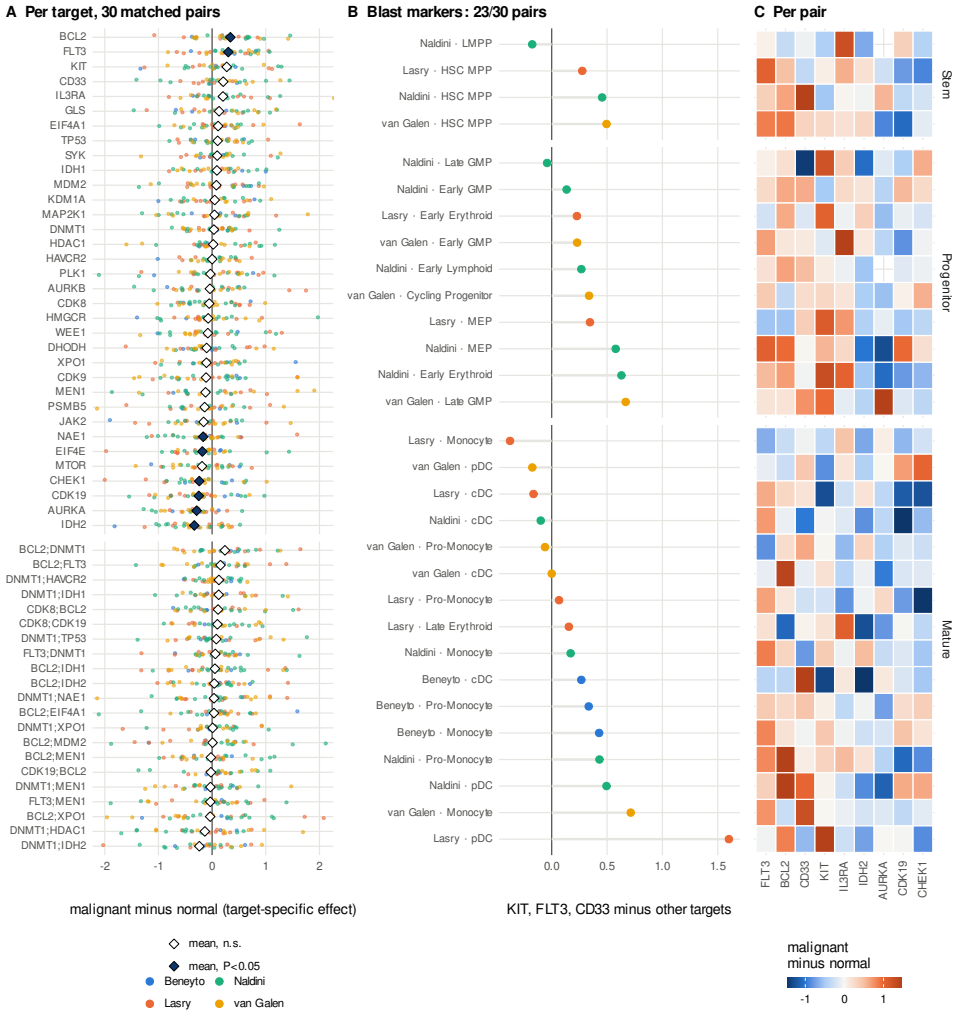


Fig. 4 Malignant identities against their normal counterparts. Each of the 30 matched pairs is a malignant identity (*-like*) and the normal identity of the same broad cell type in the same cohort; the difference is taken on the target-specific effect, so identity offsets cancel. (A) Malignant minus normal for each target over the 30 pairs (dots, coloured by cohort) and their mean (diamonds; filled, Wilcoxon signed-rank $P < 0.05$); x axis cut at the 0.5th and 99.5th percentiles. Targets are ordered by the mean, single genes above joint targets. (B) Mean malignant-minus-normal difference of the blast markers KIT, FLT3 and CD33 minus that of the other targets, per pair, grouped by compartment. (C) Malignant minus normal for selected targets in the same pairs as B; white, a pair without a score.

shows how variable single pairs are: no single target passed FDR 0.05 over the 30 pairs (lowest 0.06), and the signal emerges only when pairs, or targets with a shared role such as the blast markers, are pooled.

Joint targets are not the sum of their members

The 21 joint targets did not reproduce either member's profile across identities. Their scores correlated weakly with the stronger member (median Spearman ρ 0.22 over the 84 joint target × cohort combinations) and less with the weaker one (0.09), and they scored 2.5-fold the higher of the two members. A joint target is perturbed as one knockout of both genes' edges, and what the network then loses is not what it loses for either gene alone, so joint targets should be read as perturbations in their own right. Some of them were among the most consistent results: CDK8;BCL2 acted on stem identities in all four cohorts, and DNMT1;HAVCR2 spared them most.

Single-cohort run on the original labels

Before the four-cohort runs, we ran the pipeline on the van Galen data with the authors' own malignant/normal labels, in which every malignant cell is labelled with the normal cell type it most resembles (Fig. 5). With `--n_cells` 2,000, NetPerturb kept 19 identities (29,752 cells) and dropped pDC (225 cells) and ProB (298) automatically. In 296 of the 1,045 identity × target pairs the target had no edge in that identity's network and scored zero, most often in T and NK cells and for AURKB, CDK8, IL3RA and JAK2; in the four-cohort runs, on the current pipeline, only 13 pairs had no score.

A malignant identity scored higher than its normal counterpart in 58% of 193 comparisons (median +0.10 $\log_{10}$), most clearly at the top of the hierarchy: HSC-like exceeded HSC by 0.38 $\log_{10}$ (16 of 22 targets, signed-rank $P = 0.02$) and Prog-like exceeded Prog by 0.36 (19 of 28, $P = 0.009$; Fig. 5B). The blast markers carried the largest excess: over the three progenitor pairs, KIT, FLT3 and CD33 had a median malignant excess of 0.90 $\log_{10}$, against 0.13 for the other targets ($P = 0.006$). BCL2 ranked plasma cells first and HSC-like cells, the population venetoclax acts on [22], second. Identity offsets spanned 0.47 $\log_{10}$ (Fig. 5C), and the low offsets of normal HSC and Prog accounted for much of the general malignant excess: once both offsets were removed, the pooled malignant-minus-normal difference was −0.03 $\log_{10}$. The four-cohort runs explained these offsets by network size and confirmed that the blast-marker excess is not one of them. [TODO: This run predates the mito/ribo removal and the Data quality section (25 Sep). Re-run it on the current pipeline, or move it to an Additional file, before submission.]

Cost

A cohort took 15–43 h of compute, almost all of it GENIE3: on average 0.4–2.7 h per identity, and up to 4.7 h for an identity at the 3,000-cell cap, so the cost grows with the number of cells kept per identity. Scoring then took 4–6 min per target. Every identity and every target is its own task, so on a cluster a cohort finishes in about the time of its longest task, under 5 h.

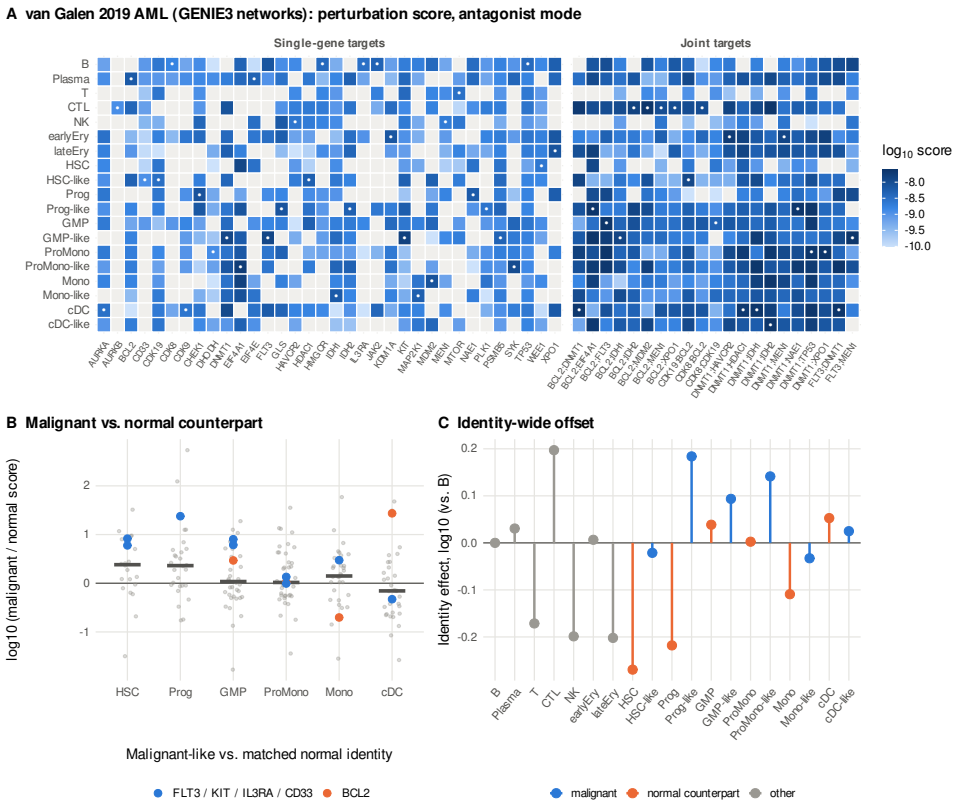


Fig. 5 Single-cohort run on the original van Galen labels [18] (GENIE3 networks, antagonist mode, seed 1). (A) $\log_{10}$ perturbation score for all 55 targets, single genes on the left and joint targets on the right, across the 19 identities, with each malignant identity (*-like*) placed under its normal counterpart. A white dot marks the top identity for each target, and grey a pair without a score (the target has no edge in that identity's network). (B) Difference in $\log_{10}$ score between each malignant identity and its normal counterpart, one point per target scored in both; bars show medians; the blast-marker targets and BCL2 are coloured. (C) Identity main effect from a decomposition of $\log_{10}$ scores into target and identity effects, relative to B cells.

Discussion

NetPerturb takes an analyst from a Seurat object to a shareable report on GRN-based in silico perturbation with a single command. It runs several network inference methods and a virtual knockout under one interface, and its seeding, containers and provenance make each result reproducible.

The four-cohort benchmark also shows how these scores should be read. Target-level results replicated between cohorts, but single identity × target scores did not, so conclusions should rest on effects aggregated over identities and, where possible, over cohorts. Scores carry an identity-wide offset that tracks how many cells an identity's network was built from. Identities should therefore be compared through the target-specific part of the score, or against control targets from the same run, and not through raw scores. Scores follow how strongly an identity's network relies on the target, which tracks the target's expression there rather than a drug's clinical sensitivity, and they are comparable only within one inference method.

Read this way, the scores agreed with what is known of AML stem cells. IL3RA [20] and KDM1A [21], two leukaemic stem cell dependencies, acted on malignant stem identities, and mitotic kinases spared stem cells, as expected of a largely quiescent compartment. Targets that acted mainly on normal HSC/MPP, such as PSMB5, CDK8 and HMGCR, are candidates for haematopoietic toxicity. BCL2 shows the limit of a network score: venetoclax eliminates leukaemic stem cells by disrupting their energy metabolism [22], yet BCL2 was not stem-specific here, because the score measures how much a network relies on the gene, not how a drug acts.

The benchmark has limits. The malignant/normal call agreed with curated labels for 81% of cells on the held-out cohort and is in-sample for the other three. Only GENIE3 networks and one seed were run across cohorts. Targets whose drugs act on a mutant protein, such as IDH1 and IDH2, cannot be read from expression. The knockout track was not evaluated here; it ships with a random-knockout null model for the same reason the offsets above exist.

Conclusions

NetPerturb makes GRN-based in silico perturbation of scRNA-seq data reproducible and scalable, from a processed Seurat object to a single report. Across four AML cohorts, its target-level scores replicated, and once identity-wide offsets were removed they separated targets that act on malignant stem cells, on normal stem cells, or spare them. [TODO: Future work: e.g. downsampling every identity to the same number of cells, which would remove most of the offset; built-in control targets and a per-identity null; detection filter on targets; AnnData input; nf-core submission.]

Availability and requirements

Project name: NetPerturb
Project home page: https://github.com/MCBLab/NetPerturb
Operating system(s): Linux and macOS (any system that runs Nextflow), on a workstation, an HPC scheduler or the cloud

Programming language: Nextflow (DSL2), R
Other requirements: Nextflow $\geq$ 22.10.1 (with Java), and Docker or Singularity
License: [TODO: MIT – the repository has no LICENSE file yet]
Any restrictions to use by non-academics: [TODO: None]

Abbreviations

AML, acute myeloid leukaemia; AUC, area under the receiver operating characteristic curve; FDR, false discovery rate; GMP, granulocyte–monocyte progenitor; GRN, gene regulatory network; HPC, high-performance computing; HSC/MPP, haematopoietic stem cell/multipotent progenitor; KO, knockout; LMPP, lymphoid-primed multipotent progenitor; MEP, megakaryocyte–erythroid progenitor; PCR, principal component regression; scRNA-seq, single-cell RNA sequencing; TOM, topological overlap matrix; WT, wild type.

Supplementary information. [TODO: Additional file 1: screenshots of the NetPerturb HTML report. Additional file 2: per-target effects of the four-cohort benchmark, stem specificity and malignant-minus-normal statistics (`target_effects.tsv`). Additional file 3: the HTML reports of the four cohort runs. Additional file 4: archive of the NetPerturb source code at the submitted release.]

Declarations

Ethics approval and consent to participate. Not applicable. This study reanalyses published, de-identified scRNA-seq data; ethics approval and consent were obtained by the original studies.

Consent for publication. Not applicable.

Availability of data and materials. The source code, container recipes and test data definitions are available at https://github.com/MCBLab/NetPerturb (release [TODO: vX.Y.Z], archived at Zenodo [TODO: DOI]). The multi-cohort benchmark uses the processed objects of Zeng et al. [14] (Zenodo, doi:10.5281/zenodo.18528442) for the cohorts of Beneyto-Calabuig et al. [15], Lasry et al. [16], Naldini et al. [17] and van Galen et al. [18] (GEO accession GSE116256), with malignant/normal calls from scMalignantFinder. The re-annotated Seurat objects, the retrained classifier and the benchmark outputs are available at [TODO: Zenodo DOI].

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. [TODO: e.g. D.M.-C.: Conceptualization, Software, Writing – original draft. . . . All authors read and approved the final manuscript.]

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